

MeriCloset

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Chapter 1

Introduction

The product described in this document is an online clothing system designed to address the challenges facing Pakistani consumers in their clothing shopping experiences. The document is to be read by developers, marketing staff, project managers, users, and testers. [1].

1.1 Problem Statement

We are developing this system to address the challenges faced by Pakistani consumers in their clothing shopping experiences. The current market is fragmented, with numerous online and physical stores offering a wide range of products. This fragmentation makes it difficult for consumers to compare options, discover new brands, and efficiently complete their purchases. The shopping experience is thus often time-consuming, frustrating, and sub-optimal. There is a need for a unified platform that brings together Pakistan's diverse fashion offerings, providing consumers with a seamless and convenient shopping experience. The proposed online clothing system is the prototype for a centralized platform to fulfill this vision.

1.2 Scope

The scope of the proposed online clothing platform encompasses the development and implementation of a centralized and personalized shopping solution for Pakistani consumers. The platform will offer a curated selection tailored to individual preferences through recommendation algorithms. Users will be able to easily browse, compare and purchase items, mainly various styles of shalwar, kameez, pants, and shirts. To enhance the shopping experience, a chatbot will be integrated to provide assistance and answer

user queries. The platform will also prioritize data privacy and security, implementing robust measures to protect user information. We believe that this project will serve as a prototype for a product to provide much needed support to local businesses and promote the growth of the Pakistani fashion industry by providing a platform for domestic brands to reach a wider audience.

1.3 Modules

1.3.1 User Management

Handle user registration, login, profile management, and password recovery.

1. User registration and login
2. Profile creation and management (preferences, saved items, order history)
3. Password recovery and security features

1.3.2 Product Catalog

Manage the product database, provide search functionality and provide personalized recommendations.

1. Product database (product details, images, categories, brands)
2. Search functionality (keyword search, filtering by category, price, brand)
3. Product recommendations based on user preferences and behavior

1.3.3 Shopping Cart

Enable users to add items to their cart, view and manage their purchases, and apply discounts.

1. Add to cart functionality
2. Product quantity adjustment
3. View and manage cart items

1.3.4 Recommendation Engine

Utilize algorithms to suggest products tailored to individual user preferences.

1. Collaborative filtering algorithms
2. Content-based filtering
3. Hybrid recommendation approach

1.3.5 Administration

Manage the back-end operations of the platform, including the product catalog, orders, and user accounts.

1. Managing product catalog and inventory
2. Monitoring platform performance and analytics
3. Managing user accounts and permissions

1.3.6 Lookbook

Manage user-created lookbooks, allowing users to add products to lookbooks, organize outfits, and share curated collections.

1. Lookbook management (create, update, delete, and view lookbooks)
2. Add/remove products to/from lookbooks
3. Allow public and private access to lookbook
4. Allow organization by theme or occasion
5. Track user engagement with lookbooks (number of views, shares)

1.3.7 Closet

Manage user's virtual closet, allowing them to store purchased or saved items, track outfits, and manage product collections.

1. Closet management (add, remove, and organize items in user's closet)

2. Add custom notes and tags to items (e.g., "Worn at event", "Favorite piece")
3. Filter closet items by category, brand, size, or occasion

1.4 User Classes and Characteristics

Table 1.1: User Classes and Characteristics

User class	Description
Admin	An Admin is a user who has complete control over the online clothing system. Admins have the ability to add, edit, and delete brands, products, and users. They can also monitor system performance, troubleshoot technical issues, and respond to customer inquiries. Admins may also have access to analytics and reporting tools to track system usage and identify areas for improvement.
Brand	A Brand is a user who represents a clothing brand or retailer. Brands can create profiles on the online clothing system to showcase their products. They have the ability to upload product images, descriptions, and pricing information. Brands can also manage their inventory and communicate with customers.
Client	A Client is a user who wants to purchase clothing items online. There are millions of potential Customers in Pakistan, with a significant portion expected to use the online clothing system regularly. Customers will often browse through product catalogs, search for specific items, create wishlists, and leave reviews.

Chapter 2

Project Requirements

2.1 Use-case

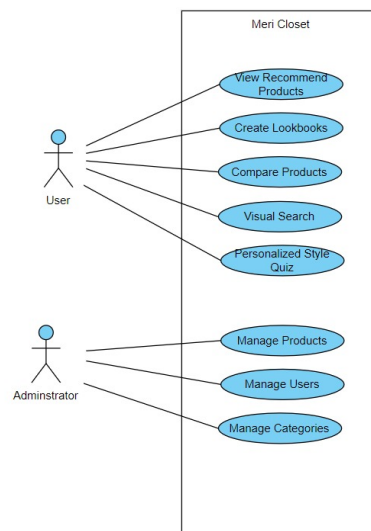


Figure 2.1: Use-Case Diagram

UC ID		
Name	Register	
Scope	MeriCloset App	
Level	User Goal	
Primary Actor	User	
Stakeholders and Interests	User: wants to create an account to personalize product recommendations and access other features.	
Pre-condition	User has access to the registration form.	
Success Guarantee	User is successfully registered and logged into their account.	
Main Success Scenario	Actor Actions	System Responsibility
	1. User clicks the "Register" button.	
		2. System displays the registration form.
	3. User enters account details (name, email, password, etc.).	
		4. System validates the input details.
	User is taken to their personalized dashboard.	
Extensions	The system will display an error if the email is already registered. The system will display an error if the password doesn't meet the requirements.	

Figure 2.2: Extended Use Case 01

UC ID		
Name	Login	
Scope	MeriCloset App	
Level	User Goal	
Primary Actor	User	
Stakeholders and Interests	User: wants to access their account and personalized features.	
Pre-condition	User has registered an account.	
Success Guarantee	User is successfully logged in and can access their account.	
Main Success Scenario	Actor Actions	System Responsibility
	1. User clicks the "Login" button.	
		2. System displays the login form.
	3. User enters credentials (email/password or social login).	
		4. System validates the credentials.
	5. User clicks the "Submit" button.	
		6. System logs the user in and redirects them to their dashboard.
Extensions	The system will display an error if the credentials are incorrect. The system will prompt for password recovery if needed.	

Figure 2.3: Extended Use Case 02

UC ID		
Name	View Recommended Products	
Scope	MeriCloset App	
Level	User Goal	
Primary Actor	User	
Stakeholders and Interests	User: wants to see product recommendations based on their preferences.	
Pre-condition	User has login an account.	
Success Guarantee	User can see personalized product recommendations.	
Main Success Scenario	Actor Actions	System Responsibility
	1. User navigates to the "Recommended Products" section.	
		2. System displays a list of products based on the user's preferences, behavior, and past interactions.
Extensions	The system will display an error if there are no recommendations available.	

Figure 2.4: Extended Use Case 03

UC ID		
Name	Visual Search (Future)	
Scope	MeriCloset App	
Level	User Goal	
Primary Actor	User	
Stakeholders and Interests	User: wants to find similar products by uploading an image.	
Pre-condition	Visual search functionality is available, and the user is logged in.	
Success Guarantee	User can see products like the uploaded image.	
Main Success Scenario	Actor Actions	System Responsibility
	1. User clicks on the "Upload Image" button.	
		2. System prompts the user to upload an image.
	3. User selects an image from their device.	
		4. System processes the image using image recognition.
	5. User submits the image.	
		6. System displays similar products from different brands.
Extensions	The system will display an error if the uploaded image is too large or in an unsupported format.	

Figure 2.5: Extended Use Case 04

UC ID		
Name	Personalized Style Quiz	
Scope	MeriCloset App	
Level	User Goal	
Primary Actor	User	
Stakeholders and Interests	User: wants to complete a quiz to personalize their style preferences for better recommendations.	
Pre-condition	User is logged in and accesses the style quiz.	
Success Guarantee	User's quiz results are saved, and personalized recommendations are provided.	
Main Success Scenario	Actor Actions	System Responsibility
	1. User starts the style quiz.	
		2. System presents questions about style, fit, color, and brand preferences.
	3. User answers quiz questions.	
		4. System saves the answers and processes the results.
	5. User submits the quiz.	
		6. System updates personalized recommendations based on the quiz.
Extensions	The system will prompt the user if any required quiz question is left unanswered.	

Figure 2.6: Extended Use Case 05

2.2 Functional Requirements

2.2.1 User Management

1. User Registration:

- The system shall allow users to register with a unique username and password.
- The system shall validate user input data (e.g., email address, phone number).

2. User Login:

- The system shall allow users to log in with their registered username and password.
- The system shall authenticate user credentials and grant access to the system.

2.2.2 Product Catalog

1. Product Database:

- The system shall store product details (e.g., description, price, category, brand).
- The system shall store product images.

2. Search Functionality:

- The system shall allow users to search for products by keyword, category, price range and brand.

3. Product Recommendations:

- The system shall provide personalized product recommendations based on user preferences (e.g., saved items, order history).
- The system shall provide personalized product recommendations based on user behavior (e.g., engagement time).

2.2.3 Administration

1. Managing Product Catalog and Inventory:

- The system shall allow administrators to add, edit, and delete products from the product catalog.

- The system shall allow administrators to manage product inventory levels and track stock quantities.
- The system shall provide real-time updates on product availability and inventory levels.

2. Managing User Accounts and Permissions:

- The system shall allow administrators to create, edit, and delete user accounts.

2.2.4 Shopping Cart

1. Add to Cart Functionality:

- The system shall allow users to add products to their shopping cart.
- The system shall validate the product quantity and availability before adding to cart.
- The system shall update the cart contents and display the updated cart summary.

2. View and Manage Cart Items:

- The system shall display the cart contents, including product details and quantities.
- The system shall allow users to remove products from their cart.
- The system shall allow users to apply discounts and promotions to their cart.
- The system shall calculate and display the subtotal, tax, and total cost of the cart contents.

2.2.5 Lookbook

1. Lookbook Management:

- The system shall allow users to create, update, and delete lookbooks.
- The system shall allow users to view their own lookbooks and those shared with them.
- The system shall provide a user-friendly interface for managing lookbooks, including adding and removing products.

2. Add/Remove Products to/from Lookbooks:

- The system shall allow users to add products to their lookbooks.

- The system shall allow users to remove products from their lookbooks.
- The system shall update the lookbook contents in real-time.

3. Public and Private Access to Lookbook:

- The system shall allow users to set their lookbooks as public or private.
- The system shall allow users to share their public lookbooks with others.
- The system shall restrict access to private lookbooks to authorized users.

4. Organization by Theme or Occasion:

- The system shall allow users to organize their lookbooks by theme or occasion.
- The system shall provide a categorization system for lookbooks (e.g., summer, winter, formal, casual).
- The system shall allow users to filter lookbooks by category.

5. Track User Engagement with Lookbooks:

- The system shall track the number of views and shares for each lookbook.
- The system shall allow users to view their own lookbook engagement metrics.

2.2.6 Closet

1. Closet Management:

- The system shall allow users to add, remove, and organize items in their virtual closet.
- The system shall provide a user-friendly interface for managing closet items, including categorization and filtering.
- The system shall update the closet contents in real-time.

2. Add Custom Notes and Tags to Items:

- The system shall allow users to add custom notes and tags to items in their closet (e.g., "Worn at event", "Favorite piece").
- The system shall provide a tagging system for closet items (e.g., categories, brands, sizes, occasions).
- The system shall allow users to filter closet items by tags.

3. Filter Closet Items by Category, Brand, Size, or Occasion:

- The system shall allow users to filter closet items by category, brand, size, and occasion, etc.

2.3 Non-Functional Requirements

2.3.1 Performance

- The response time of functions should be at most 2 seconds.
- The system should be able to handle at least 1000 concurrent users.

2.3.2 Security

- The system should encrypt user passwords using a salted hash algorithm (e.g., bcrypt) with a minimum of 10 characters.
- The system should implement rate limiting on login attempts to prevent brute-force attacks (e.g., 5 attempts within 10 minutes).
- The system should implement role-based access control (RBAC) to restrict access to administrative functions.

2.3.3 Usability

- The system should provide clear and concise error messages for invalid user input.

2.3.4 Data Integrity

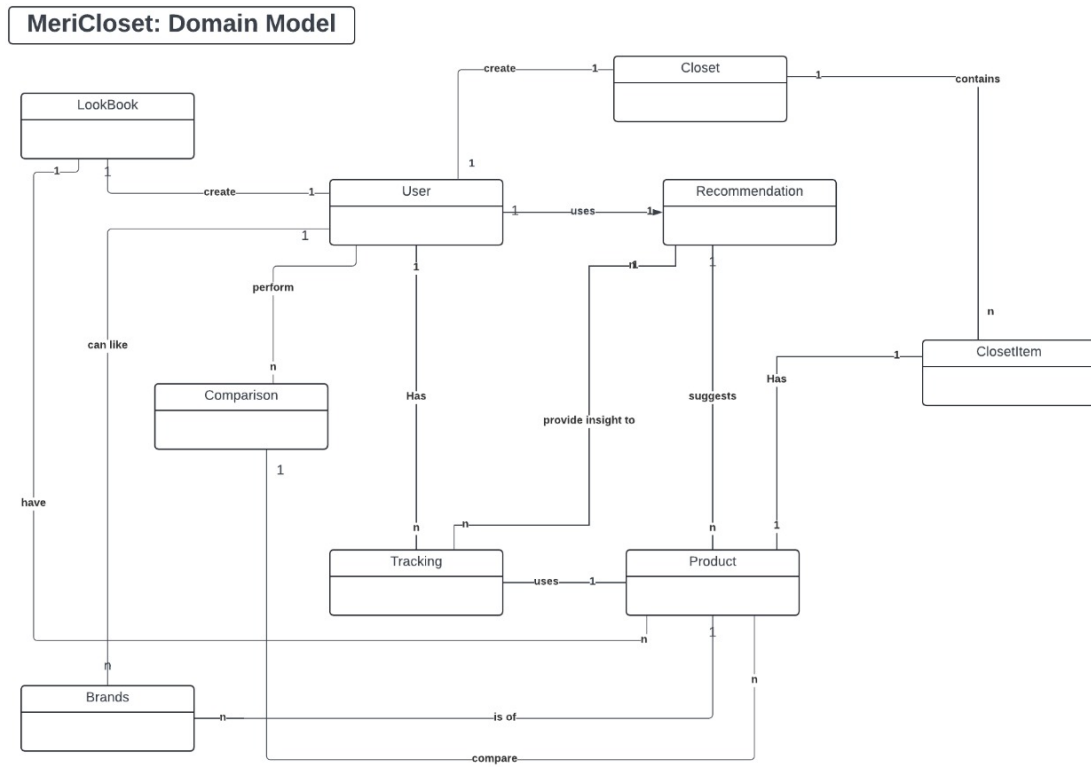
- The system should maintain a minimum of 99.5% data accuracy for product and user information (e.g., product details, images, categories, brands).
- The system should update product and user information in real-time to reflect changes in inventory or product details.

2.3.5 Accuracy

- The system should achieve a minimum of 80% accuracy in recommending products based on user preferences and behavior.
- The system should continuously learn and improve recommendation accuracy through machine learning algorithms.

2.4 Domain Model

figure



Domain Model

Chapter 3

System Overview

The proposed online clothing system is the prototype for a centralized platform to fulfill our vision of a centralised shopping system containing the items of all the brands in Pakistan that want to sell their items online. The platform will offer a curated selection tailored to individual preferences through recommendation algorithms. Users will be able to easily browse, compare and purchase items, mainly various styles of shalwar, kameez, pants, and shirts. To enhance the shopping experience, a chatbot will be integrated to provide assistance and answer user queries. The platform will also prioritize data privacy and security, implementing robust measures to protect user information.

3.1 Architectural

Design

MeriCloset is a personalized shopping app designed to enhance the online shopping experience for users by offering tailored recommendations and features. The app aims to bridge the gap between user preferences and the vast array of clothing options available online. By utilizing advanced data insights and machine learning, MeriCloset will provide users with a curated shopping experience that considers their individual tastes, body measurements, and occasions for which they are shopping.

3.2 Design

Models

Design Models for Object Oriented Development Approach We have made the following diagrams as we believe they best fit our approach for development:

- Activity Diagram
- Class Diagram

Activity Diagram

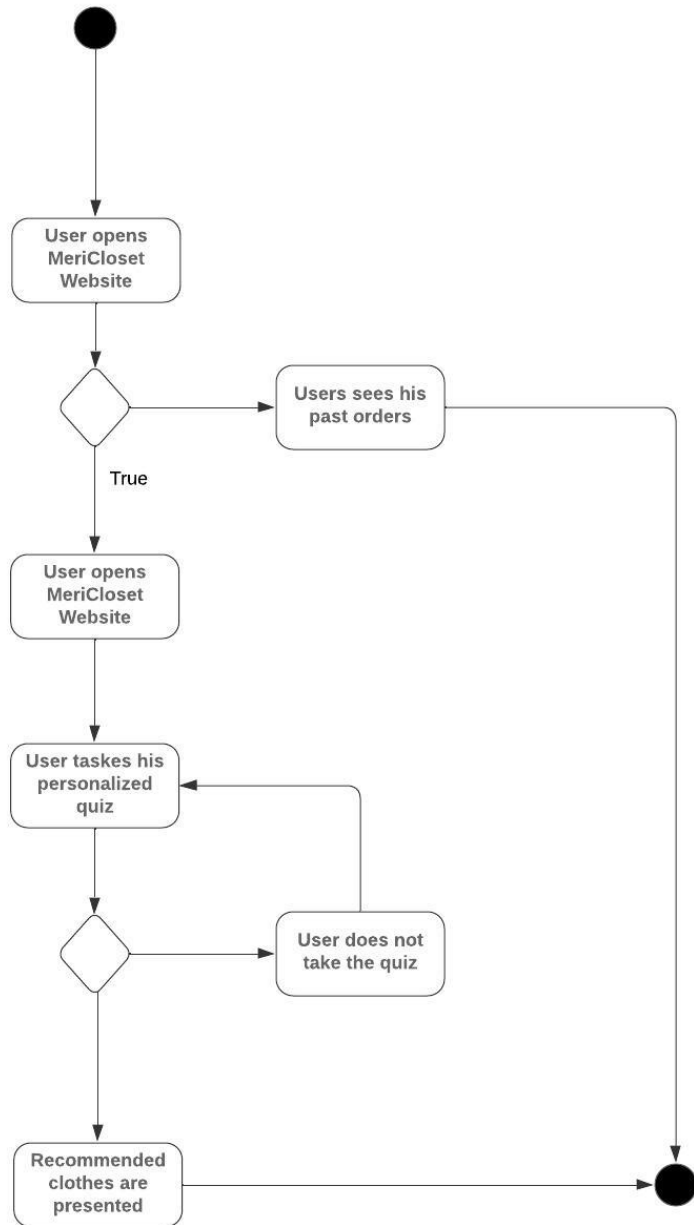


Figure 3.1: Activity Diagram

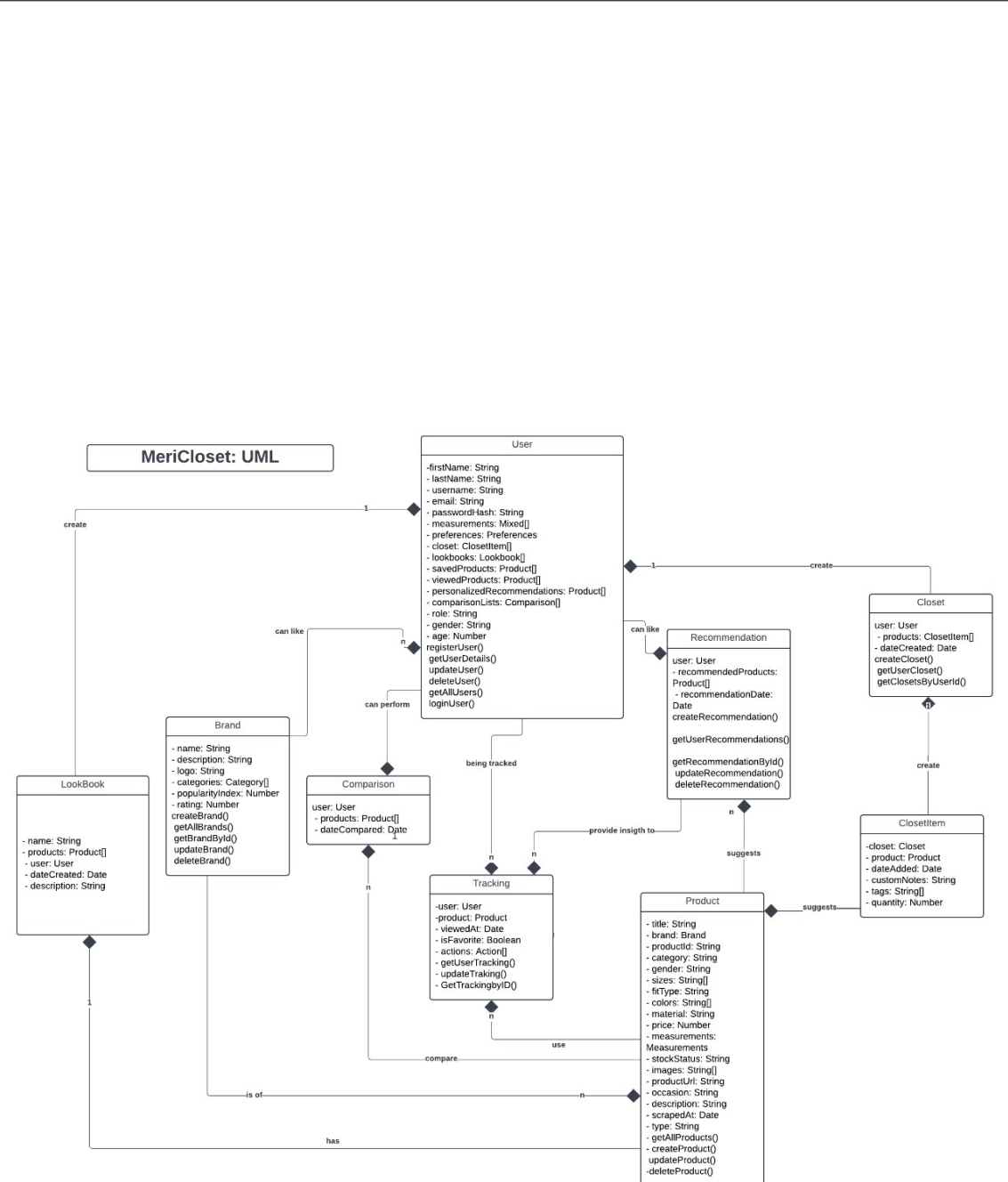


Figure 3.2: UML

3.3 Data

Design

In this project, we are working locally so everything is within a local environment. We have created a database that we have built on MongoDB as our data is very versatile due to the nature of our project. The integration of this with our application and its future deployment is also something we are adept at hence our choice.

We fetch the item details from our database to our application, the images are stored in link format and user can view the items images and the relevant details. This is something we have done from the beginning in order to not deal with the hassle of creating a file system for the purpose of documenting our progress.

Chapter 4

Conclusions and Future Work

We have created an application that works as a platform which unites and brings together Pakistan diverse fashion offerings. This is our proposed vision for a centralized platform which will provide consumers with a seamless, efficient and aggregated shopping experience. Our application provides a tailored selection for individual preferences through the use of recommendation applications. Our users can freely express their fashion sense and share it with others through lookbooks. They can store their favorite products in their personal online closets.

Our project, however, only serves as a prototype for a platform to provide greater reach for domestic brands. Some items have been taken from public datasets while others have been scrapped from publicly available web pages of local brands. Future application can be created can use this project as a jumping off point to reach greater heights. It is our hope that these applications will provide Pakistan-made clothing a chance to truly flourish.

Bibliography

- [1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.